Local stability of a sharp q-deformed commutator response for conditional quantum operations

Xiaojian Zeng
PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

The real overlap of an ordinary commutator with a q-deformed commutator has a known sharp upper bound for two-dimensional matrices and for normal matrices in arbitrary dimension. We study its optimized response near an embedded two-level saturation witness. For every fixed finite dimension $n \geq 3$ and every real $q \notin \{-1, 1\}$, we prove that the response has a Morse–Bott local maximum at the known sharp constant. Its local equality set is a smooth real manifold of dimension $4n - 2$, characterized by one complex relation on a reducing two-dimensional block. We calculate the complete normal Hessian spectrum, including perturbations that couple the active levels to additional levels, and obtain a two-sided quadratic deficit in the Frobenius distance to the equality manifold. The resulting neighborhood includes nonnormal, full-rank matrices, while the second operator remains unrestricted. An auxiliary determinant identity characterizes all nonzero two-dimensional equality matrices at $q = 1/2$. After contraction scaling, the response is a success-weighted observable of coherently ordered conditional operations on a maximally entangled input. The theorem supplies a local sharp envelope and exact normal susceptibilities for that observable. No explicit neighborhood radius, uniform limit, or resolution of the unrestricted global conjecture is claimed.

Keywords: conditional quantum operations; deformed commutator; Frobenius norm; equality manifold; local stability; operator interference.

1 Introduction

Operator products in different orders give distinct conditional amplitudes. Their squared norms and real overlap can be combined into a weighted order response. A useful matrix functional for this purpose is

$$f(A, B; q) = \operatorname{Re} \operatorname{Tr}\big((BA - AB)^\dagger (BA - qAB)\big), \qquad q \in \mathbb{R}. \tag{1}$$

Here $A, B \in M_n(\mathbb{C})$, the inner product is $\langle X, Y \rangle = \operatorname{Tr}(X^\dagger Y)$, and $\|X\|_F^2 = \operatorname{Tr}(X^\dagger X)$ is the unnormalized Frobenius norm. The parameter q is a real response weight.

Chruściński, Kimura, Ohno and Singal introduced this functional and proved the sharp estimate

$$f(A, B; q) \leq c(q) \|A\|_F^2 \|B\|_F^2, \qquad c(q) = \frac{1 + q + \sqrt{2(1 + q^2)}}{2}, \tag{2}$$

when $n = 2$, and when A is normal in any dimension [1, Theorem 1]. Their unrestricted all-dimensional statement is a conjecture in that source. We use its author-preprint version for the equations, equality witness and two-dimensional proof. The ordinary Böttcher–Wenzel point $q = 1$ and the point $q = -1$ are already known cases.

The question addressed here is local: does the sharp witness remain a maximum under arbitrary small matrix perturbations, including nonnormal perturbations and couplings to

extra levels? We answer this for every fixed finite $n \geq 3$ and every real $q \notin \{-1, 1\}$. The zero directions of the Hessian form a true curved equality manifold, rather than an unresolved higher-order instability. The remaining directions have an explicit strictly negative spectrum. Together with a simple spectral branch, this controls the full local remainder and proves a sharp local bound for all complex B.

This functional differs from the squared norm of a q-deformed commutator studied in [2, 3]. It also differs from the averaged relaxation functional used in open quantum dynamics. The universal maximal relaxation-rate bound has been proved in [4]; the present result supplies no new relaxation-rate bound. Its physical content is instead the local envelope and normal susceptibility of a success-weighted conditional-operation response. The unrestricted inequality (2) is not proved here.

2 Response operator and local stability theorem

Let column vectorization identify $M_n(\mathbb{C})$ with $\mathbb{C}^{n^2}$. Right and left multiplication by A are represented by

$$P_A = A^{\mathsf{T}} \otimes I_n, \qquad Q_A = I_n \otimes A. \tag{3}$$

Expanding (1) gives the Hermitian response operator

$$H_q(A) = P_A^\dagger P_A + q Q_A^\dagger Q_A - \frac{1+q}{2}(P_A^\dagger Q_A + Q_A^\dagger P_A). \tag{4}$$

Thus

$$\begin{aligned} f(A, B; q) &= \operatorname{vec}(B)^\dagger H_q(A) \operatorname{vec}(B), \\ g_q(A) &= \frac{\lambda_{\max}(H_q(A))}{\|A\|_F^2} = \max_{B \neq 0} \frac{f(A, B; q)}{\|A\|_F^2 \|B\|_F^2}. \end{aligned} \tag{5}$$

The function g_q is defined for $A \neq 0$, invariant under unitary conjugation, and unchanged by multiplying A by a nonzero complex scalar. No restriction on B is imposed by this optimization.

Fix $q \notin \{-1, 1\}$, and abbreviate

$$\begin{gathered} R = \sqrt{2(1+q^2)}, \quad c = c(q), \quad u = \sqrt{c-1}, \quad b = \operatorname{sgn}(1+q)\sqrt{c-q}, \\ A_q = \operatorname{diag}(u, -b, 0, \ldots, 0). \end{gathered} \tag{6}$$

Useful exact relations are

$$u^2 + b^2 = R, \qquad u^2 - b^2 = q - 1, \qquad 2ub = 1 + q, \qquad c > \max(1, q). \tag{7}$$

In particular $u, b, u - b, u + b$ are nonzero on this parameter domain.

For a complex 2×2 matrix C, set

$$\Phi_q(C) = \det C + \frac{ub}{(u-b)^2}(\operatorname{Tr} C)^2. \tag{8}$$

Near A_q, define $\mathcal{M}_q$ to be the local family of matrices supported on a reducing complex two-plane, with active block C near $\operatorname{diag}(u, -b)$ and $\Phi_q(C) = 0$. Being supported means that in an orthonormal decomposition into this plane and its complement the matrix is $C \oplus 0$. Distance to $\mathcal{M}_q$ below is distance to this local manifold in a sufficiently small tubular neighborhood. We use the real Frobenius metric $\operatorname{Re}\operatorname{Tr}(D^\dagger E)$ on matrix variations.

Theorem 1 (Local sharp envelope and normal spectrum)**.** *For every fixed finite $n \geq 3$ and every real $q \notin \{-1, 1\}$, the function g_q is real analytic near A_q and has a Morse–Bott local*

maximum c. Its local maximum set and local critical set are exactly $\mathcal{M}_q$, a smooth embedded real manifold of dimension $4n-2$. At A_q, the nonzero eigenvalues of its real Hessian are

eigenvalue	*multiplicity*
-2	2
$-a_1 = -\dfrac{4c(c-q)}{(2c-q)R}$	$2(n-2)$
$-a_2 = -\dfrac{4c(c-1)}{(2c-1)R}$	$2(n-2)$
$-a_3 = -\dfrac{2c}{R}$	$2(n-2)^2$

(9)

All four magnitudes are positive. With $\mu = \min(2, a_1, a_2, a_3)$ and $\Lambda = \max(2, a_1, a_2, a_3)$, there is a sufficiently small, n, q-dependent neighborhood U of A_q on which

$$\frac{\mu}{8}\operatorname{dist}_F(A, \mathcal{M}_q)^2 \le c - g_q(A) \le \Lambda \operatorname{dist}_F(A, \mathcal{M}_q)^2. \tag{10}$$

Consequently, for every $A \in U$ and every $B \in M_n(\mathbb{C})$,

$$f(A, B; q) \le \left[c - \frac{\mu}{8}\operatorname{dist}_F(A, \mathcal{M}_q)^2\right] \|A\|_F^2 \|B\|_F^2. \tag{11}$$

The theorem asserts existence of a neighborhood, not an explicit radius. It contains full-rank and fully nonnormal perturbations of A_q, including arbitrary couplings to the additional levels. There is no claim of uniformity as n grows or q approaches an excluded point. The distance estimate is expressed at the unnormalized witness (6); its constants refer to that choice of Frobenius coordinates.

3 Equality geometry and the largest spectral branch

Lemma 2 (Embedded equality family)**.** *The family $\mathcal{M}_q$ is smooth of real dimension $4n-2$. On a neighborhood of A_q within $\mathcal{M}_q$, one has $g_q = c$, the largest eigenvalue of H_q is simple, and the full gradient of g_q vanishes.*

Proof. The two-dimensional saturation witness of [1] admits the Schur family

$$C(w) = \begin{pmatrix} u & w \\ 0 & -b \end{pmatrix}, \qquad B(w) = \begin{pmatrix} -bw/R & 1 \\ 0 & uw/R \end{pmatrix}, \qquad w \in \mathbb{R}. \tag{12}$$

Substitution into (4), using (7), gives

$$H_q(C(w))\operatorname{vec}(B(w)) = c(R + w^2)\operatorname{vec}(B(w)), \qquad \Phi_q(C(w)) = 0. \tag{13}$$

The known two-dimensional bound makes this the largest eigenvalue. This verification uses the signed b and therefore includes $q < -1$.

The polynomial Φ_q depends only on trace and determinant. For its zeros near $\operatorname{diag}(u, -b)$, the two unordered eigenvalues are $\alpha u, -\alpha b$, with $\alpha \neq 0$. Indeed their sum is $\alpha(u-b)$, and their product is $-\alpha^2 ub$; conversely these roots give $\Phi_q = 0$. Unitary Schur reduction, a diagonal unitary and complex scaling then reduce such a block to (12) with $w \ge 0$. Thus every block in the local Φ_q family attains $g_q = c$.

At the base, differentiating $\Phi_q = 0$ gives

$$\delta C_{22} = -\frac{b}{u}\,\delta C_{11}, \qquad \delta C_{12}, \delta C_{21} \in \mathbb{C}. \tag{14}$$

The complex gradient is nonzero, so the active equality set has real dimension six. The Grassmannian of complex two-planes in $\mathbb{C}^n$ contributes $4(n-2)$ dimensions. The active block remains invertible nearby. Its supporting plane is therefore uniquely recovered as the orthogonal complement of the kernel of the full matrix. This proves local embeddedness and the stated dimension.

At A_q, the matrix units diagonalize H_q, and the top eigenvector is E_{12}, with eigenvalue cR. The gap to E_{21} is $(1-q)^2 > 0$. The other nonzero channels have eigenvalues among u^2, b^2, qu^2, qb^2, all strictly below cR. The remaining channels have eigenvalue zero. This establishes simplicity, which persists near the base.

For completeness, separation also holds on nearby embedded equality blocks. The two off-plane forms are right multiplication with response $\|BC\|_F^2$ and left multiplication with response $q\|CB\|_F^2$. The first has top eigenvalue at most $\|C\|_F^2$. For $q \geq 0$, the second has top eigenvalue at most $q\|C\|_F^2$; for $q < 0$, it is negative semidefinite and its top eigenvalue is at most zero. Both are strictly below $c\|C\|_F^2$. The active largest branch remains the simple one found above.

Active-block derivatives of g_q vanish because the known two-dimensional bound is attained there. For off-plane and bottom-block variations, the first variations of BA and AB, evaluated at an active maximizing B, are orthogonal to the active products. The first variation of $\|A\|_F^2$ also vanishes in these directions. The simple-eigenvalue derivative therefore makes the full gradient zero. Unitary invariance extends this argument to every nearby supporting plane. □

4 Complete normal Hessian

We give the calculation in real and imaginary matrix-entry coordinates; E_{ij} and iE_{ij} form a real orthonormal basis. Write $\lambda(A) = \lambda_{\max}(H_q(A))$ on its simple branch, and $e = \mathrm{vec}(E_{12})$. For an entry direction D_i, let P_i, Q_i denote its right and left multiplication operators, and set $L_i = P_i - Q_i$, $M_i = P_i - qQ_i$. For the corresponding derivative H_i at A_q, simple-eigenvalue differentiation gives

$$\lambda_{ij} = \mathrm{Re}\big(\langle L_i e, M_j e\rangle + \langle L_j e, M_i e\rangle\big) + 2\,\mathrm{Re}\sum_{r\neq 12} \frac{(H_i)_{12,r}(H_j)_{r,12}}{cR - (H_q(A_q))_{r,r}}, \tag{15}$$

$$(g_q)_{ij} = \frac{\lambda_{ij} - 2c\delta_{ij}}{R}. \tag{16}$$

The last formula uses stationarity at A_q, $\lambda = c\|A_q\|_F^2$, and the Euclidean Hessian of the squared Frobenius norm.

On the active real diagonal coordinates the Hessian block is

$$-\frac{2}{R}\begin{pmatrix} b^2 & ub \\ ub & u^2 \end{pmatrix}. \tag{17}$$

The imaginary block is identical. Each has eigenvalues $0, -2$; the zero direction is $(u, -b)$, consistent with (14). The four real active off-diagonal directions are tangents to the equality family and have zero Hessian.

Let $\alpha \geq 3$ be a spectator index. For the real pair $(A_{1\alpha}, A_{\alpha 1})$, the first-derivative vectors $H_i e$ point only along $E_{\alpha 2}$, with coefficients

$$\frac{(1+q)b + 2qu}{2}, \qquad \frac{(1+q)b}{2}. \tag{18}$$

The branch gap is

$$cR - b^2 = u^2(2c - q) > 0. \tag{19}$$

Using $(1+q)b + 2qu = 2uc$ in (15)–(16) yields

$$-\frac{a_1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{20}$$

For the real pair $(A_{2\alpha}, A_{\alpha 2})$, the derivative vectors point only along $E_{1\alpha}$, with coefficients

$$-\frac{(1+q)u}{2}, \qquad -\frac{2b+(1+q)u}{2}. \tag{21}$$

Now the gap and simplifying relation are

$$cR - qu^2 = b^2(2c-1) > 0, \qquad 2b + (1+q)u = 2bc. \tag{22}$$

The result is (20) with a_1 replaced by a_2. For each imaginary pair the off-diagonal signs reverse, so the nonzero eigenvalue is unchanged. These zero directions are precisely the unitary variations of the supporting two-plane.

For a direction D wholly within the spectator block, $DE_{12} = E_{12}D = 0$, and $H'_D e = 0$. Only the norm correction in (16) survives, giving $-2c/R = -a_3$ for each real component. There are $2(n-2)^2$ such components.

There are no mixed Hessian blocks. Distinct spectator indices couple to orthogonal matrix units; the two active-spectator sectors couple respectively to $E_{\alpha 2}$ and $E_{1\alpha}$; active derivatives have only active support. The direct-product cross terms in (15) and the norm cross terms vanish between these sectors as well. This support argument holds for every finite n. It is not an inference from a finite list of computed dimensions.

The spectrum is therefore (9). Since $c > \max(1, q)$, $R > 0$, $2c - q > 0$ and $2c - 1 > 0$, all the displayed normal magnitudes are strictly positive. The negative rank and nullity are

$$2 + 4(n-2) + 2(n-2)^2 = 2(n-1)^2, \qquad 2n^2 - 2(n-1)^2 = 4n - 2. \tag{23}$$

Stationarity on $\mathcal{M}_q$ puts its tangent space in the Hessian kernel. Equality of dimensions identifies that kernel exactly.

5 Control of the full local remainder

Completion of the proof of Theorem 1. The simple largest eigenvalue of a real-analytic Hermitian matrix family is real analytic locally. Since $A_q \neq 0$, the quotient g_q has the same property. In a sufficiently small tubular neighborhood write $A = p + v$, where $p \in \mathcal{M}_q$ and v is Frobenius-normal to $\mathcal{M}_q$ at p. Then $\operatorname{dist}_F(A, \mathcal{M}_q) = \|v\|_F$. Lemma 2 gives $g_q(p) = c$ and $dg_q(p) = 0$.

At the base, the Hessian of $h = c - g_q$ is positive on the normal space, with smallest and largest eigenvalues μ, Λ. Continuity of the Hessian and of the normal bundle permits shrinking the neighborhood so that, for every such normal segment and $0 \le t \le 1$,

$$\frac{\mu}{4}\|v\|_F^2 \le D^2h(p+tv)[v,v] \le 2\Lambda\|v\|_F^2. \tag{24}$$

Here v is normal at p; it need not be normal to a different tangent plane at $p + tv$. Taylor's integral identity is exact:

$$h(p+v) = \int_0^1 (1-t)D^2h(p+tv)[v,v]\,dt. \tag{25}$$

It proves (10). The radial derivative $dh(p+v)[v]$ is strictly positive whenever $v \neq 0$, by the corresponding integral without the factor $1 - t$. There are therefore no other nearby critical points. The equality manifold is exactly the local maximum set, and the Hessian is nondegenerate in its normal directions: this is the Morse–Bott assertion. Finally, (5) proves (11) for every B, with $B = 0$ trivial. □

The argument controls the entire analytic remainder. A negative quadratic truncation alone would not establish a bound along the Hessian kernel; the exact equality manifold is essential. The theorem also excludes a sequence of violating pairs whose first operators approach A_q at fixed n, q, even if the second operators vary arbitrarily.

6 Conditional-operation observable

For nonzero matrices A, B, choose positive scalars $\alpha = 1/(2\|A\|_F)$ and $\beta = 1/(2\|B\|_F)$. The operators $K_A = \alpha A$ and $K_B = \beta B$ are interior contractions, since their operator norms are at most $1/2$. Each is a legal success Kraus operator; adjoining $(I - K^\dagger K)^{1/2}$ completes its two-outcome trace-preserving instrument.

Use the normalized maximally entangled input

$$|\Phi_n\rangle = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} |j\rangle|j\rangle. \tag{26}$$

With $Y = K_B K_A$, $X = K_A K_B$, define the successful ordered amplitudes $|\psi_Y\rangle = (Y \otimes I)|\Phi_n\rangle$ and $|\psi_X\rangle = (X \otimes I)|\Phi_n\rangle$. The exact trace identity

$$\langle \psi_Y, \psi_X \rangle = \frac{\mathrm{Tr}(Y^\dagger X)}{n} \tag{27}$$

fixes the normalization. Assume coherent control of these two orders and retain the joint-success branch. Its unnormalized state is

$$|\Omega\rangle = \frac{|0\rangle|\psi_Y\rangle + |1\rangle|\psi_X\rangle}{\sqrt{2}}. \tag{28}$$

The reduced, subnormalized control state ρ obeys

$$\rho_{00} = \frac{\|Y\|_F^2}{2n}, \quad \rho_{11} = \frac{\|X\|_F^2}{2n}, \quad \mathrm{Tr}(\sigma_x \rho) = \frac{\mathrm{Re}\,\mathrm{Tr}(Y^\dagger X)}{n}. \tag{29}$$

Thus the Hermitian control observable

$$O_q = 2|0\rangle\langle 0| + 2q|1\rangle\langle 1| - (1+q)\sigma_x \tag{30}$$

has success-weighted response

$$\mathcal{R}_q = \mathrm{Tr}(O_q \rho) = \frac{f(K_A, K_B; q)}{n}. \tag{31}$$

Failure events contribute zero to this response. An expectation in the normalized postselected state would instead be $\mathcal{R}_q / \mathrm{Tr}\,\rho$, when the success probability is nonzero, and is a different observable. Negative q is a signed measurement weight, not a negative probability or transition rate.

For A in Theorem 1's neighborhood, homogeneity transfers its bound to the physical operators:

$$\mathcal{R}_q \le \frac{\|K_A\|_F^2 \|K_B\|_F^2}{n} \left[c - \frac{\mu}{8} \,\mathrm{dist}_F(A, \mathcal{M}_q)^2 \right]. \tag{32}$$

The distance in this expression is measured in the original representative A, before contraction scaling. The normalized response ratio is unchanged by that scaling. At the saturation witness, an infinitesimal perturbation in a unit Frobenius normal eigenvector with Hessian eigenvalue $-a$ has optimized deficit $c - g_q(A_q + tD) = at^2/2 + O(t^3)$. Equation (9) therefore gives exact normal susceptibilities, including both kinds of couplings to extra levels. No particular implementation, success efficiency or experimental error radius is predicted.

7 Comparison, limitations and reproducibility

The established two-dimensional bound supplies the active equality family. The advance here is its stability in an open, full-dimensional neighborhood in $M_n(\mathbb{C})$, the geometry of all local maximizing first operators, and the complete normal response spectrum. The auxiliary two-dimensional equality identity in Appendix A is an extraction of the known two-dimensional inequality and is not a separate novelty claim.

Known bounds for $\|AB - qBA\|_F^2$ [2, 3] concern a different response. For example, the immediate combination of the ordinary commutator bound with the general q-norm bound by Cauchy–Schwarz gives $\sqrt{3}$ at $q = 1/2$, whereas the sharp response constant is $(3+\sqrt{10})/4 < \sqrt{3}$. This simple implication is insufficient; we do not claim that every possible transformation of prior results has been exhausted. Likewise, the proved universal relaxation bound [4] is not this unaveraged all-q response problem. A bounded primary-literature comparison identified no matching local theorem; it is not exhaustive priority certification.

The result is finite-dimensional and dimensionless. There is no continuum limit, boundary condition, hidden perturbative approximation or finite-size extrapolation. The excluded values $q = -1, 1$ have known global bounds but do not satisfy the nondegenerate geometry used here. No explicit radius is obtained, and the theorem does not settle (2) for general finite displacements. Determining an effective radius or extending control beyond the local equality family would require further work.

7.1 Symbolic checks and use of AI tools

The analytic argument is accompanied by exact SymPy 1.14.0 checks in Python 3.12. The supplied programs construct (4) from multiplication operators, verify the Schur equality pair, and reduce rational polynomial residuals modulo $u^2 - b^2 - 2ub + 2 = 0$. Separate programs check the determinant identity in the appendix and complete real-coordinate Hessians at $q = 1/2$ in dimensions three and four. The every-dimensional conclusion follows from the matrix-unit support argument above, not from those finite computations. The checks use exact identities, not floating-point fitting or an enclosure inferred from a scan. Their environment and regeneration instructions accompany the local draft.

OpenAI Codex was used in literature screening, proposing and developing arguments, implementing and executing symbolic calculations, checking the mathematics and operator interpretation, and drafting this manuscript. The automated consistency checks do not constitute external peer review or completed human scientific review. Human author review and approval of this local draft remain pending.

A Complete two-dimensional equality criterion at $q = 1/2$

Proposition 3. *Let $q = 1/2$ and $c = (3+\sqrt{10})/4$. For every nonzero $A \in M_2(\mathbb{C})$, $g_{1/2}(A) = c$ if and only if*

$$\det A + \frac{3}{2}(3+\sqrt{10})(\operatorname{Tr} A)^2 = 0. \tag{33}$$

Proof. Put $D = c\|A\|_F^2 I_4 - H_{1/2}(A)$ and $\Phi(A) = \det A + \frac{3}{2}(3+\sqrt{10})(\operatorname{Tr} A)^2$. The known two-dimensional theorem makes D positive semidefinite. Direct determinant expansion yields

$$\det D = \frac{\|A\|_F^4}{64}|\Phi(A)|^2. \tag{34}$$

To perform the expansion for a general matrix, use unitary Schur form $A = \left(\begin{smallmatrix} a & t \\ 0 & d \end{smallmatrix}\right)$, where

$a, d \in \mathbb{C}$ and $t \geq 0$. In the column-vectorized basis the response matrix is

$$\begin{pmatrix} t^2 & -\overline{a}t/4 & t(d-3a/4) & -3t^2/4 \\ -at/4 & h_{22} & 0 & td/4 \\ t(\overline{d}-3\overline{a}/4) & 0 & h_{33} & t(\overline{a}/2-3\overline{d}/4) \\ -3t^2/4 & t\overline{d}/4 & t(a/2-3d/4) & t^2/2 \end{pmatrix}, \tag{35}$$

with

$$h_{22} = |a|^2 + \tfrac{1}{2}|d|^2 + \tfrac{3}{2}t^2 - \tfrac{3}{2}\,\mathrm{Re}(\overline{a}d),$$
$$h_{33} = |d|^2 + \tfrac{1}{2}|a|^2 - \tfrac{3}{2}\,\mathrm{Re}(\overline{a}d).$$

Subtraction from $c(|a|^2 + |d|^2 + t^2)I_4$ and expansion, with $(\sqrt{10})^2 = 10$, gives (34). Both sides are invariant under unitary conjugation, so Schur reduction proves it for every complex A. Positive semidefiniteness and $A \neq 0$ imply that its vanishing is equivalent to the largest response eigenvalue being $c\|A\|_F^2$. This includes nonzero nilpotent matrices. □

Statements and Declarations

Data and code availability. No experimental data are used. Exact verification programs and reproduction instructions are included in the local handoff materials. A public archival location has not yet been assigned.

Funding and competing interests. These declarations require confirmation by the human author before submission.

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References

[1] Dariusz Chruściński, Gen Kimura, Hiromichi Ohno, and Tanmay Singal. One-parameter generalization of the Böttcher–Wenzel inequality and its application to open quantum dynamics. *Linear Algebra and its Applications*, 656:158–166, 2023. doi: 10.1016/j.laa.2022.09.022. URL `https://arxiv.org/abs/2208.10005v1`. Equations and proofs are cited from author preprint arXiv:2208.10005v1 (2022), titled One parameter generalization of BW inequality and its application to open quantum dynamics.

[2] Dariusz Chruściński, Gen Kimura, Hiromichi Ohno, and Tanmay Singal. Bounding the Frobenius norm of a q-deformed commutator. Author preprint, arXiv:2202.11520v2, 2022. URL `https://arxiv.org/abs/2202.11520v2`.

[3] Motoyuki Nobori. A note on some upper bounds for the Frobenius norm of the q-deformed commutator. Author preprint, arXiv:2608.03897v1, 2026. URL `https://arxiv.org/abs/2608.03897v1`.

[4] Paolo Muratore-Ginanneschi, Gen Kimura, and Dariusz Chruściński. Universal bound on the relaxation rates for quantum Markovian dynamics. Author preprint, arXiv:2409.00436v1, 2024. URL `https://arxiv.org/abs/2409.00436v1`.